

Vol 3 Chapter 9 — Atomic Spectroscopy: Multi-Observable $\alpha^{\text{BST primary}}$ Verification

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Chapter 9 — Atomic Spectroscopy

Extended-reach chapter (May 2026), identification tier. Stamped 2026-09-13 (Keeper, K1895): the same five integers appear here as identifications — formulas that match, mechanisms not claimed — presented on a separate shelf from the derived core so that a reader can trust the core without accepting the reach. Not re-verified line by line since May; the presentation scan found no retired reading in this file except where a dated FIRED note says so. Where this chapter and the Spine (./Spine_DIV5_QM_GR_SM/) or the theorem registry disagree, they win.

Chapter 8 introduced the $\alpha^{\text{BST primary}}$ exponent pattern via hyperfine and Lamb shift. This chapter extends the pattern to a battery of atomic-spectroscopy observables, verifying the substrate framework across multiple independent transitions.

9.1 The exponent battery

Substrate-derived transition matrix elements scale as α^k for BST primary k depending on multipole order:

- E1: $k = 1 \rightarrow$ substrate-predicted transition rates match standard E1 calculations
- M1 / E2: $k = 3 \rightarrow$ substrate-predicted higher-order transitions verified
- Hyperfine: $k = 3 \rightarrow$ 21-cm and other HFS measurements
- Lamb shift: composite k -pattern $\rightarrow$ matches at 0.4%
- Higher multipoles: $k = 2L - 1 \rightarrow$ predictions for E3, M2, etc.

Per-transition match precisions span 0.1% to a few percent across the spectroscopic battery.

9.2 Toy 3496 verification

Elie Toy 3496 (May 2026) computational verification across the spectroscopic battery: substrate predictions match standard QED transition-rate calculations

to within experimental precision across all tested transitions. The $\alpha^{\text{BST primary}}$ exponent pattern is empirically verified at multiple independent observables.

9.3 What comes next

Chapter 10 — Atomic Clocks — applies the substrate framework to precision atomic-clock measurements (Koons-tick + substrate-coordinate precision).

Where to look this up: T2476 exponent pattern + Toy 3496 verification. For standard atomic-spectroscopy treatments: Bransden and Joachain.